Generative AI in court

GENERATIVE AI IN COURT

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ABSTRACT

In the age of burgeoning AI-driven creativity, the song "Heart on My Sleeve" stands out, attributing its sound to famed artists like Drake and The Weeknd without their participation. This product of 'Generative AI' is a testament to AI's creative capabilities. However, amidst the advancement of AI in the creative domain, the intersection of technology and copyright law is witnessing unprecedented challenges. Legal realms are ablaze with contentious cases. Getty Images' legal battle with Stability AI, accused of unlawfully harnessing copyrighted images for their AI art tool Stable Diffusion, underscores this tension. Parallelly, artists like Sarah Andersen, Kelly McKernan, and Karla Ortiz have taken on AI solution providers, alleging misuse of copyrighted content. Sarah Silverman and fellow authors have similarly contested OpenAI and Meta's practices. The justification at the heart of these case is two-fold: the rightsholders' intent to preserve their control over evolving technology that repurposes existing content, and the disruptive potential of AI in reshaping a traditionally human-centred creative landscape. This collision spotlights global courts grappling with AI's challenges to conventional copyright concepts. As AI challenges conventional notions of authorship and copyright, critical questions arise: Can AI infringe copyright through its learning process? Can AI be an author and own a copyright? Can AI-generated outputs violate copyrights? This chapter delves into these pressing issues pending before multiple courts, contextualizing them within the global landscape of Generative AI and its potential disruptions to the creative industry.

INTRODUCTION

The spring sensation, "Heart on My Sleeve," has taken the world by storm. Though it bears the unmistakable hallmarks of superstars Drake and The Weeknd, neither artist contributed to the song's creation. Instead, the track emerged from the power of artificial intelligence, mimicking their iconic voices with uncanny accuracy. Launched by an anonymous TikTok creator, its rapid viral ascent has resulted in millions of streams, yet also ignited significant copyright concerns. Universal Music Group, representing The Weeknd, has voiced strong opposition to such AI-driven musical creations, challenging the ethics and legality of the technology.[1] This is not an isolated dispute but rather the forefront of an escalating debate over intellectual property rights in the AI era.[2] The surge in AI's capabilities in the realm of creativity is remarkable, underscored by the rise

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[1] See e.g. Ben Schreckinger, "AI vs the culture Industry", *Politico*, April 24, 2023, https://www.politico.com/newsletters/digital-future-daily/2023/04/24/ai-vs-the-culture-industry-00093550.

[2] See "Master List of lawsuits v. AI, ChatGPT, OpenAI, Microsoft, Meta, Midjourney & other AI cos.", (14 July 2023, https://chatgptiseatingtheworld.com/2023/07/14/master-list-of-lawsuits-v-ai-chatgpt-openai-microsoft-meta-midjourney-other-ai-cos/?fbclid=IwAR3vMv9MIZRWvv1xRcponMYRjo0De7BmOJqiUc7dhn50RPBtI_0-2wqFT3I.

of 'Generative AI'.[3] Such systems have shown prowess in penning poems, stories, and news pieces; crafting melodies; retouching photos; devising video games; and producing paintings and other artistic visuals. The recent introduction of expansive text-to-image, text-to-music, and text-to-text models, including the likes of Dall-E, Stable Diffusion, Midjourney, Jasper, Chat GPT-3, and MusicLM, signifies a pivotal moment in AI's progress, underscoring the immense possibilities in AI-fueled creativity. Chat GPT-3, an advanced linguistic generative model, generates responses indistinguishable from human interaction based on textual cues. Dall-E, named in homage to both WALL-E and Salvador Dalí, crafts visual artworks from succinct text prompts.[4] Jasper operates akin to an adept copywriter, spinning content akin to seasoned writers from mere headlines.[5] Innovation like Google's MusicLM, which orchestrates intricate music pieces from textual inputs, and ventures like Meta's make-a-video and Google's Imagen video, are constantly evolving.[6]

In this context, both cultural and legal spheres are now witnessing clashes between artists, media giants, AI developers, and the tech platforms that host them. Recently, Getty Images has taken legal action against Stability AI, the makers of the AI art tool Stable Diffusion, asserting that the firm unlawfully utilized and processed countless copyrighted images to hone its software.[7] In a separate landmark lawsuit, artists such as Sarah Andersen, Kelly McKernan, and Karla Ortiz initiated an unprecedented class action for copyright infringement against AI solution providers including Stability AI Ltd., Midjourney Inc., and DeviantArt Inc. They contend these entities used vast amounts of copyrighted images without acquiring permissions or providing payment to inform their AI applications.[8] Furthermore, Sarah Silverman, along with other writers, have filed a lawsuit against OpenAI and Meta, asserting that AI training breached their copyrights, particularly when enabling users to produce summaries of their literary works.[9] Meanwhile, the U.S. copyright

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[3] See Proposal for a regulation of the European Parliament and of the Council on laying down harmonised rules on artificial intelligence (Artificial Intelligence Act) and amending certain Union legislative acts, COM(2021)0206 – C9-0146/2021 – 2021/0106(COD) (hereafter 'AI Act'), art 28 b 4 (characterizing this as "foundation models powering AI systems, designed to autonomously create content ranging from text and images to audio and video.")

[4] See Casey Newton, "How DALL-E could power a creative revolution", *The Verge*, 10 June 2022, https://www.theverge.com/23162454/openai-dall-e-image-generation-tool-creative-revolution; OpenAI, "DALL·E: Creating Images from Text", *OpenAI Blog*, 5 January 2021), https://openai.com/blog/dall-e.

[5] See Tom Simonite, 'The Future of the Web Is Marketing Copy Generated by Algorithms' (*WIRED*, April 18 2022) <https://www.wired.com/story/ai-generated-marketing-content.

[6] See Andrea Agostinelli et al, "MusicLM: Generating Music From Text@, Google Research (2023) https://research.google/pubs/pub52118; Alex Wilkins, "How will AIs that generate videos from text transform media online?", *NewScientist* November 15, 2022, https://www.newscientist.com/article/2346939-how-will-ais-that-generate-videos-from-text-transform-media-online.

[7] See James Vincent, "Getty Images is suing Stability AI, creators of popular AI art tool Stable Diffusion, over alleged copyright violation", *The Verge*, April 2, 2023, https://www.theverge.com.

[8] See *Andersen et al v. Stability AI Ltd. et al Andersen et al v. Stability AI Ltd. et al*, Docket No. 3:23-cv-00201 (ND Cal, 13 January 2023) (Complaint), https://www.courtlistener.com/docket/66732129/andersen-v-stability-ai-ltd. See also Karla Ortiz Testimony, Hearing of the U.S. Senate Committee on the Judiciary, Subcommittee on Intellectual Property (12 July 2023), https://www.judiciary.senate.gov/imo/media/doc/2023-07-12_pm_-_testimony_-_ortiz.pdf (written testimony), https://www.youtube.com/watch?v=uoCJun7gkbA (video of U.S. Senate subcommittee hearing).

[9] See *Sarah Silverman v. OpenAI, Inc.*, U.S. District Court for the Northern District of California, No. 3:23-cv-03223 (filed 7 July 2023) (proposed class action) <https://www.courtlistener.com/docket/67569254/silverman-v-openai-inc>. See also *Paul Tremblay and Mona Awad v. Open AI, Inc.*, U.S. District Court for the Northern District of California, No. […] (filed 28 June 2023), https://copyrightlately.com/pdfviewer/tremblay-v-openai-class-action-complaint/?auto_viewer=true#page=&zoom=auto&pagemode=none.

office, the District Court of Columbia and other courts in multiple jurisdictions have been asked to review the protectability of AI-generated works under the present copyright framework.[10]

The justification at the heart of these case is two-fold. On one hand, copyright holders seek to maintain their influence over rapidly advancing technologies, questioning the extent of their control over new tools that repurpose existing content. In addition, given AI's transformative potential that might impact the landscape of human labor and disrupt the human role in the creative sector,[11] these stakeholders desire safeguards to preserve their market share. Consequently, the integration of traditional copyright systems with AI-driven innovation is attracting significant judicial scrutiny globally.[12] AI's capability to generate content challenges established principles of copyright, authorship, and ownership, prompting pivotal debates over the legitimacy of protecting AI-produced works. Beyond the dilemma of recognizing AI as an 'author', courts are grappling also with two other intertwined issues: the potential of AI to violate copyright during its learning phase, and its capacity to produce content that may infringe existing copyrights.[13] This chapter aims to thoroughly explore these pressing concerns, which are currently—and will continue to be—presented before courts in multiple jurisdictions, emphasizing their global relevance in the era of Generative AI and AI-generated creativity.

1. RATIONAL: HOLD-OUT POWERS AND MARKET DISRUPTION

As we embark on an exploration of the burgeoning legal landscape surrounding Generative AI, two core justifications for the mounting lawsuits against Generative AI platforms become evident. Firstly, copyright holders are striving to retain their foothold in an environment dominated by rapidly progressing technologies. Their primary concern centres around the extent to which they can exert control over these novel tools, particularly when these tools have the capability to repurpose existing content in unprecedented ways. Secondly, the undeniable potential of AI to reshape the dynamics of human labour cannot be overlooked. Human creators yearn for mechanisms to safeguard their market positions against the sweeping tide of machine-driven creativity. This part of the chapter sets the stage for understanding these motivations, which are driving the wave of litigation against Generative AI platforms.

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[10] See infra fn. 113-120.

[11] See e.g. Tyna Eloundou and others, "GPTs are GPTs: An Early Look at the Labor Market Impact Potential of Large Language Models", arXiv:2303.10130 [econ.GN] (2023), https://arxiv.org/abs/2303.10130 (finding that findings reveal that around 80% of the U.S. workforce could have at least 10% of their work tasks affected by the introduction of LLMs, while approximately 19% of workers may see at least 50% of their tasks impacted).

[12] See Maria Iglesias Portela, Sharon Shamuilia and Amanda Anderberg, "Intellectual Property and Artificial Intelligence: A Literature Review", Joint Research Centre, Publications Office of the European Union (2019).

[13] Please note that his chapter reworks and develops content and conclusions discussed in earlier pieces. See Giancarlo Frosio, "Should We Ban Generative AI, Incentivise it or Make it a Medium for Inclusive Creativity?", in *A Research Agenda for EU Copyright Law*, eds. Enrico Bonadio and Caterina Sganga (Edward Elgar, forthcoming 2023); Giancarlo Frosio, "The Artificial Creatives: the Rise of Combinatorial Creativity from Dall-E to GPT-3", in *Handbook of Artificial Intelligence at Work: Interconnections and Policy Implications*, eds. Martha Garcia-Murillo, Ian MacInnes, and Andrea Renda (Edward Elgar, forthcoming 2023); Giancarlo Frosio, "Four Theories in Search of an A(I)uthor" in *Research Handbook on Intellectual Property and Artificial Intelligence*, ed. Ryan Abbott (Edward Elgar 2022), 155-177.

1.1. Hold Out Powers

In the wave of recent lawsuits against generative AI platforms, we observe a recurring theme: copyright law wielded as a strategic instrument to secure or enhance market dominance. Historically, the allure of potential hefty statutory damages, especially in the United States, has often made litigation a more enticing route than forging business partnerships. This preference is further bolstered by the inherent unpredictability of fair use decisions, which can deter newcomers, sparking battles over technological innovations and, in turn, encouraging monopolistic tendencies.[14] This raises important concerns about the historical struggle between copyright and innovation. This tension has consistently played out in courts, shaping the landscape of creative industries and determining the extent to which emerging technologies can thrive.

The annals of copyright law reveal a persistent tug-of-war between preservation and progress. Throughout history, property owners, particularly within the realm of copyright, have leveraged their hold-out power to obstruct progress.[15] From early confrontations involving piano rolls,[16] tape recorders,[17] radios,[18] and VHS recorders,[19] to more contemporary disputes over cable television,[20] peer-to-peer software,[21] mp3 music format,[22] search engines, digital thumbnails,[23] and cloud music-storage services, copyright holders have often resisted new technological frontiers. While some services like MP3Tunes successfully defended their innovative models,[24] others, like ReDigi and Tom Kabinet,[25] found themselves quashed under legal scrutiny, illustrating the ongoing friction between copyright proprietors and innovators. The cases recently started against Generative Ai platforms continue illustrating this ongoing tension.

This historical stance isn't solely about defending rights; it underscores a tactical approach where copyright protection is used to dictate the pace and direction of innovation. The murky waters of fair use and fair dealing decisions, compounded by the varying interpretations of exceptions and limitations, embolden copyright holders in their litigious pursuits. Such a posture can stymie the

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[14] See Brian Johnston, "Rethinking Copyright's Treatment Of New Technology: Strategic Obsolescence As A Catalyst For Interest Group Compromise", NYU Ann Surv Am L 64 (2008): 165, fn136; Robin Moore, "Fair Use and Innovation Policy" NYU L Rev 82 (2007): 944.

[15] See Carol Rose, 'The Comedy of the Commons: Custom, Commerce, and Inherently Public Property", *U Chi L. Rev.* 53 (1986): 711, 749-50, 752 (discussing how large public projects such as highways or railroads are vulnerable to the hold-out power of single property owners).

[16] See *White-Smith Music Publishing Company v Apollo Company* 209 US 1 (1908).

[17] See *Amstrad Consumer Electronics v The British Phonographic Industry* [1988] UKHL 15.

[18] See *Twentieth Century Music Corp v Aiken*, 422 US 151, *151* (1975).

[19] See *Sony Corp of Am v Universal Studios* 464 US 417, 429 (1984).

[20] See *Fortnightly Corp v United Artists* 392 US 390 (1968).

[21] See *Metro-Goldwyn-Mayer Studios v Grokster* 545 US 913 (2005); *A&M Records v Napster* 239 F3d 1004 (9th Cir 2001).

[22] See *UMG Recordings v MP3com* 92 F Supp 2d 349 (SDNY 2000).

[23] See eg *Perfect 10 v Amazon.com* 508 F3d 1146 (9th Cir 2007); *Vorschaubilder*, I ZR 69/08 (Bundesgerichtshof, 29 April 2010) (ruling that Google Image Search does not Infringe copyright).

[24] See *Capitol Records v MP3Tunes* 07 Civ 9931 (SDNY 2011).

[25] See *Capitol Records v ReDigi* 934 F Supp 2d 640 (SDNY 2013); 910 F3d 649 (2d Cir 2018); Case C-263/18 *NUV v Tom Kabinet* EU:C:2019:1111, para 72. See also Reto M Hilty, 'Exhaustion in the digital age' in Irene Calboli and Edward Lee, *Research Handbook on Intellectual Property Exhaustion and Parallel Imports* (Edward Elgar 2016) 64, 72 ff.

birth and growth of novel technologies and business blueprints, including generative AI. The risk? A potential market gap where innovations that might enrich the value of copyrighted content are stifled.[26] Furthermore, it hints at an evolving narrative where copyright law transitions from merely regulating copying to orchestrating technological design itself.[27]

Dominant players in the market have consistently resisted tech evolution to safeguard their market share.[28] In several episodes, these stakeholders wielded their influence to capture a slice of the economic boons ushered in by technology, sometimes regardless of any economic loss that traditional business models may have suffered as a consequence of innovation. The Google Books case serves as a poignant illustration. Here, a project that could have been a win-win for all parties—rightsholders, innovators, and the public—was mired in litigation as stakeholders vied for a piece of the revenue pie. Cases like these underscore the potential pitfalls of aggressive copyright enforcement, which might thwart projects that can enrich culture, reward innovation, and serve public interest. In a landmark decision, the US Court of Appeals for the 2nd Circuit validated the legitimacy of the original relational database.[29] Yet, despite the vast number of books cataloged, only select excerpts are accessible. Moreover, this constrained fair use argument might not withstand legal scrutiny in many European courts. For instance, a French case highlighted the challenges facing projects centered on book digitization for text and data mining.[30] A parallel can be drawn with the Sony case. At its height, Jack Valenti, the MPAA President at the time, dramatically told Congress, "the VCR is to the American film producer and the American public as the Boston Strangler is to the woman home alone."[31] However, history tells a different tale. The video rental sector significantly bolstered the film industry's growth. In both scenarios, innovative technologies, which held promise for copyright owners, faced opposition due to perceived threats they posed to the copyrighted content market.

Reflecting on these historical precedents, it's evident that the struggle between copyright and innovation persists in the realm of generative AI. As in the past, current lawsuits might see rightsholders leveraging their influence to stymie AI's transformative potential, including its ability to engender fresh modes of expression. Yet, the unparalleled prowess of AI necessitates a renewed focus on safeguarding human creativity. Navigating this balance between stimulating innovation and incentivizing creativity remains a central challenge in copyright and innovation discourse, a debate that spans over two centuries.

1.2. Creative Market Disruption

Legal challenges against AI-generated creativity are on the rise due to concerns about its potential upheaval of the creative industry. Pioneers in AI-generated music, like Tencent and Google's MusicLM, are prompting debates over AI potentially replacing human composers, especially in

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[26] See Fred von Lohmann, "Fair Use as Innovation Policy", *Berkeley Tech L J* 23 (2008): 829 (noting that fair use for private non-transformative uses should function as an 'innovation policy' by incentivizing innovators to develop new technologies that enable such copying).

[27] See e.g., Tarleton Gillespie, *Wired Shut: Copyright and the Shape of Digital Technology* (MIT Press, 2007).

[28] See e.g., *White-Smith*, 1; *Amstrad*, 159; *Aiken*, 151; *Sony*, 417; *Fortnightly*, 390; *Grokster*, 913; *Napster*, 1004.

[29] See *The Authors Guild v Google* 804 F3d 202 (2nd Cir 2015).

[30] See *Editions du Seuil et autres/Google Inc. et France* (TGI Paris, 18 December 2009).

[31] Home Recording of Copyrighted Works: Hearings Before the Subcomm on Courts, Civil Liberties, and the Administration of Justice of the Committee on the Judiciary, 97th Cong 97 (1982) (testimony of Jack Valenti), http://cryptomeorg/hrcw-hearhtm.

fields like video games and films.[32] While there's some evidence of AI's role in ideation within architecture and design,[33] substantial proof of its broad-scale adoption remains elusive. Additionally, while AI demonstrates potential in tasks like debugging and code-writing, it's uncertain whether tools like ChatGPT and AlphaCode will entirely eclipse human programmers.[34] Presently, AI creativity tools often necessitate significant human involvement in prompt engineering, data fine-tuning, and model integration.[35] As of today, these tools mainly augment rather than replace creative professionals, acting as supplementary aids rather than direct competitors. However, the discourse around AI in creativity reveals a bifurcation. Some view AI as a tool to bolster artistic productivity,[36] while others, termed neo-Luddites, resist its integration.[37] The looming concern isn't that AI will entirely supplant human artistry but that it might provide cost-effective alternatives, appealing to the mass market and satisfactory for a majority of consumers. Cases in point include Netflix's use of AI for background art in animations[38] and platforms like Soundraw and ShutterStock offering vast AI-generated music libraries.[39] Rachel Hill, leader of the Association of Illustrators, acknowledges AI's appeal for budget-conscious art directors.[40] Yet, she underscores the irreplaceable value of human illustrators, especially in conceptual stages. This distinction, however, could wane as AI evolves, potentially reducing the need for human involvement in creative processes.

Recently, issues have been raised by AI apps like Dall-E, Stable Diffusion, and Stability AI generating outputs in the style of famous artist. For example, a user in the Stable Diffusion community, complaining about the scarcity of 2D illustration styles, utilized 32 illustrations from Hollie Mengert's portfolio, a well-known illustrator and character designer based in Los Angeles, to train Stable Diffusion, replicating her unique style. The outcome was then shared under an open license for public use, with her name being used to identify the generated art style; for instance,

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[32] See Abner Li, "Google's MusicLM is rather good at creating music from text descriptions", *9to5Google*, January 28, 2023, https://9to5google.com/2023/01/28/google-musiclm.

[33] See e.g. Ben Dreith, "How AI software will change architecture and design", *Dezeen*, November 16, 2022, https://www.dezeen.com/2022/11/16/ai-design-architecture-product; Kevin Roose, "A.I.-Generated Art Is Already Transforming Creative Work", *NYTimes*, October 21, 2022, https://www.nytimes.com/2022/10/21/technology/ai-generated-art-jobs-dall-e-2.html.

[34] Davide Castelvecchi, "Are ChatGPT and AlphaCode going to replace programmers?", *Nature*, December 8, 2022, https://www.nature.com/articles/d41586-022-04383-z.

[35] See Vasili Shynkarenka, "Don't dismiss GPT-3/Stable Diffusion use cases if they don't work right away", *Vasili Shynkarenka Blog*, December 2, 2022, https://vasilishynkarenka.com/dont-dismiss-ideas-for-gpt-3-stable-diffusion-use-cases-if-they-dont-work-right-away.

[36] See Will Knight, "When AI Makes Art, Humans Supply the Creative Spark", *Wired*, July 13, 2022, https://www.wired.com/story/when-ai-makes-art.

[37] See Rich Johnston, "Comic Book Creators React To AI Artificial Intelligence Art Explosion", *Bleeding Cool*, December 6, 2022, https://bleedingcool.com/comics/comic-book-creators-react-to-ai-artificial-intelligence-art-explosion; Matt Growcoot, "Midjourney Founder Admits to Using a 'Hundred Million' Images Without Consent", *PetaPixel*, December 21, 2022, https://petapixel.com/2022/12/21/midjourny-founder-admits-to-using-a-hundred-million-images-without-consent.

[38] See Benj Edwards, "Netflix stirs fears by using AI-assisted background art in short anime film", *ArsTechnica*, January 2, 2023, https://arstechnica.com/information-technology/2023/02/netflix-taps-ai-image-synthesis-for-background-art-in-the-dog-and-the-boy.

[39] See Soundraw, https://soundraw.io; ShutterStock, https://www.shutterstock.com/music/search?artist=amper-music.

[40] "We are Training AI Twice as Fast this Year as Last. IEEE Spectrum", *ENGTalks*, June 30, 2022, https://www.enggtalks.com/news/217757/we-re-training-ai-twice-as-fast-this-year-as-last?c=2859.

by prompting "illustration of a princess in the forest, holliemengert artstyle".[41] In Stable Diffusion concept library available on the website Hugging Face alone, there are roughly a thousand models like that trained on Mengert's work. These range from styles inspired by classic and contemporary Disney animations to those influenced by the likes of Tron: Legacy and Kurzgesagt videos.[42] Creating and sharing these models is cost-effective and easily accessible. While Generative AI platforms may introduce mechanisms to avoid replicating a specific artist or author's style,[43] users often discover workarounds to such restrictions. As we will delve into later, it remains a matter of debate whether copyright law can address these uses. This is mainly because such uses could be considered transformative at their core, and the existing idea-expression dichotomy might preclude protection of style. Regardless, such developments are bound to disrupt the creative market, potentially leading to legal grievances from disgruntled creatives.

In summation, Future AI advancements might elevate the standards for top-tier creative roles, potentially sidelining many and consolidating the revenue stream from low-skilled creativity in the hands of a select few. AI's potential to significantly reshape the creative sectors, particularly in graphic design, music, and copywriting, cannot be overlooked. The rapid and potent evolution of AI underscores that industry disruptions are imminent. The ability of AI algorithms to generate high-quality creative outputs quickly and cost-effectively poses a challenge to human creatives. One key factor that amplifies the potential impact of AI is its exponential growth, both in terms of capabilities and the speed at which it learns. This 'singular' nature of AI development implies that disruption in the creative job market is not a matter of if, but when.

2. CLAIMS: LEARNING, AUTHORING AND INFRINGING

As we venture further into the age of generative AI, the established paradigms of our Intellectual Property (IP) systems, encompassing copyright, trade secrets, and patent law, face unprecedented challenges. While our current IP frameworks can secure protections for the software underpinning AI technologies[44] or computer-assisted creativity—which remains copyrightable provided the human involvement is original, [45] they grapple with the delineation of rights concerning the input used by AI and outputs produced by such AI, where minimal "non-expressive" human interaction prompts a machine to conjure its own unique expressions.[46] To effectively unpack the evolving legal landscape of AI-driven creativity, courts must address three pivotal concerns. Beyond identifying the AI as an author or "A(I)uthor", we must tackle the issues surrounding the machine as both a "Learner" and a potential "(A)Infringer". These concerns delve into the machine's

[41] See Andy Baio, "Invasive Diffusion: How one unwilling illustrator found herself turned into an AI model", *Waxy*, November 1, 2022, https://waxy.org/2022/11/invasive-diffusion-how-one-unwilling-illustrator-found-herself-turned-into-an-ai-model.

[42] ibid.

[43] See infra fn. 80.

[44] See Nathan Calvin and Jade Leung, "Who owns artificial intelligence? A preliminary analysis of corporate intellectual property strategies and why they matter", University of Oxford's Future of Humanity Institute (2020), https://www.fhi.ox.ac.uk/wp-content/uploads/Patents_-FHI-Working-Paper-Final-.pdf.

[45] See e.g., Robert Denicola, "Ex Machina: Copyright Protection for Computer-Generated Works", *Rutgers University Law Review* 69 (2016): 269-270; Robert Clark and Shane Smyth (1997), *Intellectual Property Law in Ireland*, Butterworths, Dublin; *Payer Components South Africa Ltd v Bovic Gaskins* [1995] 33 IPR 407.

[46] A related scenario emerges in the case of works co- or jointly authored, rather than assisted, by human and artificial intelligence.

capacity to potentially infringe upon existing copyrighted works during its learning process, as well as its ability to produce outputs that might infringe on copyrights. Both issues ultimately hinge on the potential attribution of copyright infringement liabilities to entities that train AI for creative generation or those employing AI tools to either assist their own creative process or to generate creativity independently.

In this section, we shall embark on a comprehensive exploration of these pressing issues, dissecting the intricate interplay between generative AI and the copyright frameworks, informed by various judicial reviews and interpretations.

2.1. The Machine Learner

A fundamental aspect of Machine Learning (ML) and various other AI processes is the necessity for input data that fuels AI learning and development.[47] This reliance on data and extensive data processing is intrinsic to the essence of ML,[48] giving rise to significant questions concerning ownership and legal protections. On one hand, the issue of data ownership emerges as a crucial consideration. The development of AI and ML systems typically requires the utilization of large datasets to enable the continuous refinement of the system's decision-making capabilities. This leads to a pressing question: Who holds ownership over the datasets employed to train the system? Furthermore, should Intellectual Property Rights (IPR) be extended to the intermediate data generated by AI/ML during the training process? On the other hand, and more central to the pending lawsuits concerning Generative AI, a vital question surfaces regarding the potential copyright infringement that could arise through machine learning. These lawsuits arise inter alia from a growing conflict between AI firms and content creators over the use of human-created images as training data. Many AI firms argue that scraping images from the web for training purposes is covered under laws like the US fair use doctrine, while rights holders contend that this constitutes a copyright violation. Our discussion will subsequently focus on unpacking this complex issue, shedding light on the legal intricacies that surround it.

The legal framework concerning the intersection of machine learning, data mining, and intellectual property rights presents a complicated landscape.[49] Specifically, there's a tension between the

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[47] See e.g., Olivier Gerber, "Artificial Intelligence and Machine Learning - Background Paper", 50th EPRA Meeting, Athens, 2019, https://www.epra.org/attachments/athens-plenary-2-artificial-intelligence-machine-learning-background-paper

[48] See e.g., Andrea Ottolia, *Big Data e Innovazione Computazionale* (Giappichelli 2017), 163 ff.

[49] See e.g., Rossana Ducato and Alain Strowel, "Limitations to Text and Data Mining and Consumer Empowerment: Making the Case for a Right to ‘‘Machine Legibility'", *International Review of Intellectual Property and Competition Law* 50(3) (2019): 649-684; Christophe Geiger, Giancarlo Frosio and Oleksander Bulayenko, 'Text and Data Mining: Art. 3 and 4 of the Directive 790/2019/EU" in *Los Derechos de autor en el mercado unico digital Europeo*, eds. Concepción Sáiz García and Raquel Evangelio Llorca (Tirant lo Banch 2019), 27 ff; Bernt Hugenholtz, "The New Copyright Directive: Text and Data Mining (Articles 3 and 4)", *Kluwer Copyright Blog*, 2019, http://copyrightblog.kluweriplaw.com/2019/07/24/the-new-copyright-directive-text-and-data-mining-articles-3-and-4; Eleonora Rosati, "Copyright as an Obstacle or an Enabler? A European Perspective on Text and Data Mining and Its Role in the Development of AI Creativity", *Asia Pacific Law Review*, 27 (2019): 198-217; Andrea Toth, "Algorithmic copyright enforcement and AI: Issues and potential solutions, through the lens of text and data mining", *Masaryk University Journal of Law and Technology* 13(2) (2019): 361-387; Christophe Geiger, Giancarlo Frosio and Oleksander Bulayenko, "Text and Data Mining in the Proposed Copyright Reform: Making the EU Ready for an Age of Big Data?" IIC 49(7) (2018): 814-844; Marco Caspers, and Lucie Guibault, *Baseline report of policies and barriers of TDM in Europe* (FutureTDM, 2016), https://project.futuretdm.eu/wp-content/uploads/2017/05/FutureTDM_D3.3-

lawful utilization of Text-and-Data-Mining (TDM) techniques and intellectual property protection. Machine learning techniques, described as algorithms trained to detect patterns for achieving specific goals,[50] rely heavily on TDM. TDM, is a method to extract information from vast digital data through automated software tools,[51] involving stages like (1) identification of materials, (2) copying and pre-processing,[52] (3) data extraction, and (4) recombination to identify final patterns.[53] The friction between IP protection and TDM arises from the fundamental principle that copyright protects creative form, not the underlying data or information.[54] Essentially, TDM shouldn't interfere with exclusive IPRs. However, certain activities in TDM may infringe on rights provided by copyright and database protection.

2.1.1. Tension between Copyright Protection and Machine Learning

TDM typically necessitates some level of copying protected subject matter, a process that might bring legal challenges, especially in certain jurisdictions such as the European Union, where even the limited copying of an excerpt might be perceived as infringing upon the right of reproduction.[55] However, a distinction exists when TDM tools are designed in such a way that they involve only minimal copying, perhaps just a few words, or crawl through data while processing each item individually. Such actions are more defensible, as they steer clear of subject matters protected either by copyright or sui generis rights, thereby reducing the risk of legal liability.[56] Contrarily, any act of reproduction within TDM activities that results in the creation of a copy of protected work may instigate copyright infringement. Several stages within the TDM process can potentially give rise to legal challenges:

(1) Pre-Processing Stage: Standardizing materials into machine-readable formats might provoke an infringement on the right of reproduction.[57] Further, the uploading of pre-processed material on

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Baseline-Report-of-Policies-and-Barriers-of-TDM-in-Europe.pdf.; Jean-Paul Triaille, Jérôme De Meeûs D'argenteuil, and Amélie De Francquen, *Study of the Legal Framework of Text and Data Mining (TDM)* (European Commission, 2014), http://ec.europa.eu/internal_market/copyright/docs/studies/1403_study2_en.pdf. An essential bibliography including literature treating this question can be found at Giancarlo Frosio, "L'(I)Autore inesistente: una tesi tecno-giuridica contro la tutela dell'opera generata dall'intelligenza artificiale" AIDA 29 (2020): 52-91, 54, fn. 13.

[50] European Commission, *White paper on Artificial Intelligence - A European approach to excellence and trust*, COM(2020) 65, 16.

[51] See e.g., Jiawei Han, Micheline Kamber, Jian Pei, *Data Mining: Concept and Techniques* (Morgan Kaufmann, 2011).

[52] In particular, copying encompasses (a) pre-processing materials by turning them into a machine-readable format compatible with the technology to be deployed for the TDM so that structured data can be extracted and (b) possibly, but not necessarily, uploading the pre-processed materials on a platform, depending on the TDM technique to be deployed.

[53] See Diane McDonald and Ursulla Kelly, "The Value and Benefit of Text Mining to UK Further and Higher Education. Digital Infrastructure", *JISC* (2012), http://bit.ly/jisc-textm; Triaille et al., *Study of the Legal Framework of Text and Data Mining (TDM)*, 28; Sholom Weiss, Nitin Indurkhya, and Tong Zhang, *Fundamentals of Predictive Text Mining, Texts in Computer Sciences* (Springer, 2010), 15.

[54] See e.g., Bernt Hugenholtz, *Auteursrecht op informatie* (Kluwer, 1989).

[55] See CJEU, C-5/08, *Infopaq International A/S v. Danske Dagblades Forening*, 16 July 2009, ECLI:EU:C:2009:465, paras 54-55.

[56] See Directive 2019/790/EU of the European Parliament and of the Council of 17 April 2019 on copyright and related rights in the Digital Single Market and amending Directives 96/9/EC and 2001/29/EC, 2019 O.J. L 130, Recital 8

[57] See Directive 2001/29/EC of the European Parliament and of the Council of 22 May 2001 on the harmonisation of certain aspects of copyright and related rights in the information society, 2000 O.J. L 167, Art. 2

a platform could violate this right, although this might be contingent upon whether the chosen TDM technique employs specific software for direct data analysis from the source.[58]

(2) Pre-Processing for Extraction: In more nuanced circumstances, TDM may also involve various alterations of a database protected by copyright, encompassing reproduction, translation, adaptation, arrangement, and other changes affecting the original selection and arrangement of the database's content.[59] For example, pre-processing for extraction might necessitate cleansing a database of irrelevant portions and data for analysis, which could violate both the right of reproduction and the right to make adaptations and arrangements.[60]

(3) Mining Stage: This crucial part of the TDM process, where data is finally extracted, may also infringe the right of reproduction, depending upon the character of the extraction and the mining software deployed.[61]

(4) Sui Generis Database Rights: TDM might further infringe upon these rights, especially regarding the extraction (and, to a lesser degree, the re-utilization) of substantial parts of a database. Here, the violation could occur even without the reproduction of original materials, as the extraction itself could infringe upon the exclusive rights bestowed upon the database owner.[62] In emphasizing the legal intricacies involved, the Court of Justice of the European Union (CJEU) has elucidated that the act of transferring data from one medium to another and its integration into a new medium is deemed an act of extraction, further complicating the landscape for TDM activities.[63]

2.1.2. The Legality of Machine Learning: Exceptions, Limitations, and Fair Uses

The legality TDM and its potential infringement on intellectual property laws hinge upon varied national approaches, exceptions, limitations, and interpretations of fair use in copyright law.[64] Exceptions, limitations, and fair uses within copyright law permit the use of copyrighted works without a license from the owner, recognizing that such usage may fulfill important public interests and fundamental rights, such as freedom of expression.[65] In the context of TDM, applying these

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[58] See Triaille et al., *Study of the Legal Framework of Text and Data Mining (TDM)*, 28.

[59] See Directive 1996/9/EC of the European Parliament and of the Council of 11 March 1996 on the legal protection of databases, 1996 O.J. L 77, Art. 5(a-b). See also See also Irini Stamatoudi, "Text and Data Mining", in *New Developments in EU and International Copyright Law*, ed. Irini Stamatoudi (Kluwer Law International, 2016), 264-265

[60] See Directive 1996/9/EC, Art. 5(a-b). See also Stamatoudi, "Text and Data Mining", 262; Triaille et al., *Study of the Legal Framework of Text and Data Mining (TDM)*, 34.

[61] See Triaille et al., *Study of the Legal Framework of Text and Data Mining (TDM)*, 31.

[62] Directive 1996/9/EC, Art. 7.

[63] Directive 1996/9/EC, Arts. 2(a), 7(1), and 7(2)(b); CJEU, C-203/02, The British Horseracing Board Ltd and Others v. William Hill Organization Ltd, 9 November 2004, ECLI:EU:C:2004:695; Stamatoudi, "Text and Data Mining", 267.

[64] See Sean Fiil-Flynn et al., "Legal reform to enhance global text and data mining research: Outdated copyright laws around the world hinder research", *Science*, 378(6623) (2022): 951-953. https://www.science.org/doi/10.1126/science.add6124.

[65] See Christophe Geiger, "Copyright's Fundamental Rights Dimension at EU Level", in *Research Handbook on the Future of EU Copyright*, ed. Estelle Derclaye (Edward Elgar, 2009), 27.

principles can be seen as promoting public interests, especially in the domains of scientific advancement and economic development.[66]

In Europe, several exceptions have been considered to shield TDM from intellectual property infringement. Notable exceptions include temporary acts of reproduction, research, private use, normal use of a database, and the extraction of insubstantial parts of a database.[67] Despite these exceptions, legal uncertainties have persisted, leading to the introduction of a specific TDM exception in the Copyright in the Digital Single Market Directive of 2019.[68] Unfortunately, this exception presents limitations; it can be opted-out by right holders or utilized without restrictions solely by public research institutions for research purposes.[69] This arrangement disadvantages private AI industries, restricting their ability to freely employ TDM for machine learning, a restriction echoed across many civil law jurisdictions.[70]

Conversely, countries following common law traditions, such as the US, Canada, UK, Australia, and New Zealand, have leveraged opening clauses or fair use models to legitimize TDM.[71] In the US, for example, TDM's application for training foundational models seems to be protected under the fair use doctrine,[72] albeit with some narrow limitations.[73] Cases like *Google Books* and *Perfect 10* have strengthened the understanding that using TDM to create search indices or generate data insights constitutes transformative fair use, thus supporting lawful TDM practices under US copyright law.[74] Moreover, landmark rulings, such as *Baker v. Selden* and subsequent reverse engineering cases,[75] further cement the legal position that TDM is likely permissible. The decision in *Baker v. Selden* emphasized that copyright protection is confined to specific expressions and not underlying ideas. Thus, protected content can be utilized and copied if it "must necessarily be used as an incident to" the use of unprotected materials.[76] This ruling can be applied to TDM for training Generative AI models, where the focus is on analysis and extraction from large datasets without replicating the original copyrighted expression. Additionally, this legal precedent implies that the lawful reproduction of protected materials may be possible when mining text and data to train Generative AI models, which themselves remain unprotected. However, courts might also consider that some authors, argue that fair use may not protect *expressive* machine learning

[66] See Joao Pedro Quintais, "Rethinking Normal Exploitation: Enabling Online Limitations in EU Copyright Law", *AMI* 6 (2017): 197-205.

[67] Geiger, Frosio, and Bulayenko, 'Text and Data Mining: Art. 3 and 4 of the Directive 790/2019/EU",31-37

[68] Directive 2019/790/EU, Art. 3 and 4.

[69] ibid

[70] See Fiil-Flynn et al., "Legal reform to enhance global text and data mining research".

[71] ibid.

[72] See eg Mark Lemley and Bryan Casey, "Fair Learning", Texas L Rev 99(4) (2021): 743, 743-785.

[73] See Henderson et al, "Foundation Models and Fair Use", preprint 1 (2023), https://arxiv.org/pdf/2303.15715.pdf (noting that 'copyrighted content may be used to build foundation models without incurring liability due to the fair use doctrine. However, there is a caveat: *If the model produces output that is similar to copyrighted data, particularly in scenarios that affect the market of that data, fair use may no longer apply to the output of the model*') (emphasis added).

[74] See *The Authors Guild*, 202; *The Authors Guild v. HathiTrust*, 755 F.3d 87 (2nd Cir. 2014); *Perfect 10 v Amazon.com*, 1146; *Perfect 10 v Google*, 416 F. Supp. 2d 828 (C.D. Cal. 2006).

[75] See e.g., *Sega Enterprises Ltd v Accolade, Inc*, 977 F. 2d 1510 (9th Cir. 1992); *Sony Computer Entertainment v Connectix Corp*, 203 F. 3d 496 (9th Cir. 2000).

[76] *Baker v. Selden*, 101 U.S. 99, 104 (1879).

applications.[77] In this context, I would propose that Sobel's distinction between non-expressive and expressive fair use doesn't compellingly align with the reality of Generative AI. The argument here pivots on the way expressions are deployed as input data for training machine learning models. When these expressions are processed for the purpose of extracting data, the foundational models at play do not interact with the protected expressions in any creative or expressive manner. Instead, the models utilize this content in a functionally analytic or non-expressive manner, serving as building blocks to construct a new, inventive machine that possesses the capability to create an endless array of original expressions. The apparent confusion in Sobel's argument likely originates from an underlying assumption that the use of input materials in crafting an expressive machine necessarily imbues that usage with an expressive character. Yet, this reasoning overlooks the subtlety that the utilization remains inherently non-expressive, corresponding more closely to precedents such as the *Google Books* case. In those instances, it's not the act of using the content that is expressive, but rather the novel machine or tool itself that manifests expressiveness. This distinction underscores the transformative nature of the process, wherein existing expressions are not simply repurposed but fundamentally reshaped, contributing to the creation of something entirely new. The challenge for courts, then, is to accurately discern the nuances of this interaction.

2.1.3. Potential policy and judicial approaches to Machine Learning

In sum, the complex relationship between TDM techniques and copyright law and the legality of data scraping for training Generative AI presents a multifaceted and jurisdiction-dependent challenge for courts. First, courts are faced with the task of navigating an intricate legal terrain where they must balance technological advancements with the protection of original creations. This task is further complicated by the rapid evolution of technology and the corresponding lag in legal adaptation. Second, the regulatory landscape's diversity in governing TDM and machine learning creates an unequal playing field, particularly for private AI industries operating within jurisdictions with more restrictive laws. While these restrictive approaches undoubtedly fortify the rights of creators and shield their market share from encroachment by AI technologies, they simultaneously impose limitations on innovation. Third, this dichotomy is starkly evident in the European context, where uncertainty surrounding the permissibility of TDM and AI training has significantly impeded the competitive edge of European AI innovators. The risk of legal liabilities within the EU, coupled with the ambiguity in interpreting and applying copyright laws to emerging technologies, has created a stifling environment. This state of inaction inadvertently tilts the balance in favor of jurisdictions like the US, where landmark cases and the application of the fair use doctrine create a more favorable ecosystem for innovation.

In addressing these challenges, courts must be guided by coherent policy considerations that recognize the multifaceted nature of the problem. Among these considerations are: encouraging innovation with policymaking geared towards fostering an environment that stimulates creativity and innovation without eroding essential copyright protections; international consistency with efforts made to create some semblance of international alignment to avoid jurisdictional disparities that may hinder global technological advancement; and ethical considerations concerning the impact of AI technologies on societal values, cultural heritage, and individual creativity must be part of the judicial deliberation.[78]

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[77] See Benjamin Sobel, "Artificial intelligence's fair use crisis", Colum. JL & Arts 41 (2017): 45, 45 (emphasis added).

[78] See Joao Pedro Quintais and Nick Diakopoulos, "A Primer and FAQ on Copyright Law and Generative AI for News Media", *Medium*, April 28, 2023, https://generative-ai-newsroom.com/a-primer-and-faq-on-copyright-law-and-

As courts grapple with these challenges, certain legal precedents and emerging technologies provide insight into how these issues might be addressed. The *Field v. Google* decision, for instance, highlights the importance of adherence to technical standards.[79] In this case, the fact that Field had not employed the "robots.txt" exclusion standard to prevent Google from web crawling his site weighed against his copyright claim of infringement by copying his content. This suggests that generative AI system developers and deployers might be well-advised to adopt technical mitigation strategies to support their fair use claims. Such strategies could align with the broader aim of fostering an environment that stimulates creativity and innovation without undermining essential copyright protections.

Meanwhile, innovative self-help solutions are emerging as a means of shielding artists from potential infringement by AI art generators. An example of this is Glaze, a technology developed at the University of Chicago, crafted to protect images from AI art generators that might replicate an artist's unique style.[80] Artists can employ Glaze to attach 'style cloaks' to their works before publishing them online. These almost imperceptible changes are intended to mislead generative AI models that might attempt to mimic the artist's style, effectively causing the AI to associate the artwork with something unrelated. To assess the efficiency and applicability of the 'glazing' process, the developers of Glaze have worked with more than 1,100 professional artists, testing it on various works. Among them is Karla Ortiz, one of the plaintiffs in a recent lawsuit against Stability AI, Midjourney, and DeviantArt. This method, however, may face challenges from future neural networks capable of recognizing and removing the 'glaze,' much like current technologies can detect and erase watermarks. Courts might conclude that developing AI that circumvents self-help solutions might weigh in favour of a copyright infringement claim.

Finally, there are legislative initiatives aimed at increasing transparency in the use of copyrighted material within Generative AI processes. Specifically, the AI Act proposed by the European Commission mandates that providers of Generative AI disclose a detailed summary of how copyrighted training data is utilized.[81] It's important to note that this disclosure requirement does not absolve the AI providers from potential legal liabilities. The proposed legislation explicitly states that this obligation exists "without prejudice to Union or national or Union legislation on copyright".[82] Nevertheless, courts may take into account whether this transparency obligation has been fulfilled when determining the final outcome of potential infringement cases. Additionally, such disclosure obligations could pave the way for creators to exercise an opt-out option, thereby establishing a market for their works to be used in training generative AI models.[83] Companies like Stability AI have already started to incorporate opt-out provisions, allowing content creators and artists to exclude their works from the next version of their product, Stable Diffusion.[84] In a

generative-ai-for-news-media-f1349f514883 (listing a number of ethical and responsible uses that should be nonetheless warranted even if generative AI models are permitted under copyright law).

[79] See *Field v. Google, Inc.*, 412 F. Supp. 2d 1106 (D. Nev. 2006).

[80] See Shawn Shan et al, "Glaze: Protecting Artists from Style Mimicry by Text-to-Image Models", arXiv:2302.04222 [cs.CR] (2023), https://arxiv.org/abs/2302.04222.

[81] See AI Act, art 28 b 4 c).

[82] ibid.

[83] See Quintais and Diakopoulos, "A Primer and FAQ on Copyright Law and Generative AI for News Media".

[84] See e.g., Benj Edwards, "Stability AI plans to let artists opt out of Stable Diffusion 3 image training", *ArsTechnica*, December 12, 2022, https://arstechnica.com/information-technology/2022/12/stability-ai-plans-to-let-artists-opt-out-of-stable-diffusion-3-image-training.

recent public statement, the CEO of Stability AI asserted that the training datasets they use are “ethically, morally, and legally sourced and used”, while acknowledging dissenting opinions by adding, “[s]ome folks disagree so we are doing opt out and alternate datasets/models that are fully cc”.[85] Courts may consider these opt-out mechanisms in their legal reasoning, possibly introducing the concept of implied licenses to adjudicate on infringement cases. As such, once these opt-out mechanisms gain widespread acceptance, they could potentially set a legal standard that influences judicial outcomes.

2.2. The A(I)uthor

The inquiry into the A(I)uthor, or the concept of a machine as an author, sets forth the pressing question of whether AI-generated creativity can be sheltered under the existing fabric of copyright law. Delving into this issue necessitates an examination of two central prerequisites for the copyright protection of creative works: authorship and originality.[86] As mentioned earlier, current intellectual property regimes, including copyright law, trade secrets, and patent law, already offer protections to the foundational software underlying AI technology. However, a distinct challenge emerges when considering the actual creative output generated by AI, for the legal safeguards afforded to the software do not readily translate to this innovative domain. A report from the European Commission candidly illustrates the dilemma, observing that the 'protection of AI-generated works [...] seems to be [...] problematic' within the existing legal framework.[87] This complexity arises, in part, from the 'humanist approach' ingrained in copyright law, leading to a contentious debate over whether AI-created works truly merit copyright protection at all.[88] In the sections that follow, we will delve deeper into the complex interplay between the notions of authorship and originality as they relate to creativity generated by artificial intelligence. We will explore why these concepts, as currently understood, may pose obstacles to extending copyright protections to AI-generated works.[89]

2.2.1. Authorship

The exploration into the copyright protectability of AI-generated creativity must first address a fundamental question: does the concept of an author, according to traditional copyright norms,

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[85] See Emad Mostaque (@EMostaque), Twitter, January 15, 2023, https://twitter.com/EMostaque/status/1614437223323111425.

[86] Alongside the matters of authorship and originality, there exists a tangential but significant question concerning the legal personality of machines generating work. This complex issue, though closely related, falls outside the scope of our discussions in this chapter. See e.g., Frosio, “Four Theories”, 159-161.

[87] Massimo Craglia et al., *Artificial Intelligence: A European Perspective* (Joint Research Centre, Publications Office of the European Union, 2018), 67, https://publications.jrc.ec.europa.eu/repository/bitstream/JRC113826/ai-flagship-report-online.pdf.

[88] ibid 68.

[89] Actually, mindful of this legal framework, Dall-E and other Generative AI programs do not claim copyright of the works generated by users via their platforms and grant users ‘full usage rights to commercialize the images they create [with DALL·E], including the right to reprint, sell, and merchandise’ See e.g. ‘New OpenAI Art Program Does not Claim Copyright for AI”, *Mind Matters*, July 22, 2022, https://mindmatters.ai/2022/07/new-openai-art-program-does-not-claim-copyright-for-ai; Sarang Sheth, “Who Owns AI-Generated Content? Understanding Ownership, Copyrighting, And How The Law Interprets AI-Generated Art”, Yanko Design, May 27, 2023, https://www.yankodesign.com/2023/05/27/who-owns-ai-generated-content-understanding-ownership-copyrighting-and-how-the-law-interprets-ai-generated-art

inherently necessitate a human creator? In other words, can an AI be recognized as an author within the legal frameworks that govern copyright?

2.2.1.1. The International Perspective

While international treaties remain largely silent on providing a concrete definition of an author, there are subtle cues within the text of the Berne Convention that may effectively bar AI from being included within the traditional understanding of an author. A couple of key points elucidate this contention. Firstly, the term of protection in copyright law, being intricately tied to the life of the author, becomes an incompatible concept when attempting to apply it to machines.[90] How could one reconcile the finite lifespan of a human with the indefinite existence of a machine? Secondly, the references made to the nationality or residence of the author within legal documents further imply a human-centric understanding.[91] This framework seems inherently skewed towards human agents, suggesting a deliberate exclusion of non-human entities from the legal definition of authorship. Cumulatively, these elements give rise to a compelling argument. As has been stated, the "humanist cast" of Berne, coupled with its underlying deference to personality theories, robustly supports the "human-centered notion of authorship presently enshrined in the Berne Convention".[92] This interpretation, by its very essence, would exclude non-human authorship—including that of AI—from the scope of Berne's protections.

2.2.1.2. The EU Perspective

In exploring the concept of a machine as an author within the European Union, the EU's stance seems to align with an anthropocentric view, where authorship is firmly rooted in human creation, even when technical aids are employed.[93] While EU statutory law doesn't provide a cross-cutting definition of authorship, significant directives and legal insights underscore the human-centric approach. Both Article 2(1) of the Software Directive and Article 4(1) of the Database Directive define an author as a natural person, a group of persons, or a legal person, reinforcing the human element in authorship.[94] The preparatory works (*travaux préparatoires*) for these directives further

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[90] Berne Convention for the Protection of Literary and Artistic Works, Art. 7

[91] ibid, Art. 3

[92] See Jane Ginsburg, "People Not Machines: Authorship and What It Means in the Berne Convention", *Int'l Rev. of Intell. Prop. and Comp. L.* 49 (2018): 131, 134-135; Sam Ricketson, "People or machines? The Berne Convention and the changing concept of authorship", *Columbia VLA Journal of Law & the Arts* 16 (1992): 1, 34. See also Tanya Aplin and Giulia Pasqualetto, "Artificial Intelligence and Copyright Protection" in *Regulating Industrial Internet Through IPR, Data Protection and Competition Law*, eds. Rosa Maria Ballardini, Petri Kuoppamäki and Olli Pitkänen, (Kluwer Law International, 2019), §5.04.

[93] See e.g., Bernt Hugenholtz and Joao Pedro Quintais, "Copyright and Artificial Creation: Does EU Copyright Law Protect AI-Assisted Output?" *IIC - International Review of Intellectual Property and Competition Law* 52 (2021): 1190 (mentioning that "although EU copyright law nowhere expressly states that copyright requires a human creator, its 'anthropocentric' focus (on human authorship) is self- evident in many aspects of the law"). See also e.g., Frosio, "Four Theories", 162-163; Jean-Marc Deltorn and Franck Macrez, "Authorship in the Age of Machine Learning and Artificial Intelligence", CEIPI Research Paper No 2018-10 (2018): 22-23, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3261329; Jean-Marc Deltorn, "Deep Creations: Intellectual Property and the Automata", *Frontiers in Digital Humanities* (2017): 8, https://www.frontiersin.org/articles/10.3389/fdigh.2017.00003/full.

[94] See Directive 2009/24/EC of the European Parliament and of the Council of 23 April 2009 on the legal protection of computer programs, O.J. L111/16; Directive 96/9/EC.

emphasize this human-centered notion, referring explicitly to "the human author who creates the work" and affirming that the natural person retains the unalienable rights to claim paternity of their work.[95] The original proposal for the Software Directive stated that despite the decreasing human input in machine-generated programs, a human "author" must always be present and possess the right to claim "authorship".[96] This viewpoint was reiterated in the Court of Justice of the European Union (CJEU) *Painer* case, where Advocate General Trstenjak stressed that protection applies solely to human creations, including those employing technical aids such as a camera. [97]

National legislation within the EU further corroborates this perspective. For instance, Article L.111-1 of the French Intellectual Property Code necessitates that a copyrightable work be a "creation of the mind";[98] Article 5 of the Spanish Copyright Act plainly defines the author as the natural person who creates the work;[99] and Article 11 of the German Copyright Act aligns authorship with a personality approach, safeguarding "the author in his intellectual and personal relationships to the work".[100] Moreover, EU law and several national legislations bolster the human-centric perspective by instituting a presumption of authorship for the *person* named in the work, barring evidence to the contrary. [101]

In sum, the EU's legal framework and jurisprudence largely support an anthropocentric vision of authorship, excluding non-human creations such as those generated by AI. This stance resonates through various legal instruments and cases, illustrating a consistent approach across the EU that appears to preclude the possibility of recognizing a machine as an author under the current copyright regime.

2.2.1.3. The US Perspective

The United States legal framework provides scant accommodation for the notion of machine or non-human authorship in the context of copyright law. While the U.S. Copyright Act does not explicitly define "author", some legal analysts have suggested that the text of the law does not necessarily restrict authorship to human beings.[102] However, both supplementary textual

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[95] See Ana Ramalho, "Will Robots Rule the (Artistic) World? A Proposed Model for the Legal Status of Creations by Artificial Intelligence Systems", 21 J. of Internet L. (2017): 12, 17-18.

[96] See European Commission, "Explanatory Memorandum to the proposal for a Software Directive" COM (88) 816 final, 21.

[97] CJEU, C-145/10, *Eva-Maria Painer v. Standard VerlagsGmbH* [2011] ECR I-12533, Opinion of the AG Trstenjak 2011, para 121

[98] Loi 92-597 du 1er juillet 1992, Code de la propriété intellectuelle, L111-1 (France) (hereafter 'French IP Code').

[99] Real Decreto Legislativo (RDL) 1/1996, de 12 de abril, por el que se aprueba el texto refundido de la Ley de Propiedad Intelectual, regularizando, aclarando y armonizando las disposiciones legales vigentes sobre la materia, BOE-A-1996-8930, art 5 (Spain) (hereafter 'Spanish IP Law).

[100] G. v. 09.09.1965, Gesetz über Urheberrecht und verwandte Schutzrechte (Urheberrechtsgesetz – UrhG), art 7 and 11 (Germany).

[101] Directive 2004/48/EC of the European Parliament and of the Council of 29 April 2004 on the enforcement of intellectual property rights, O.J. L195/16, Art. 5

[102] See Denicola, "Ex Machina", 275-283; Annemarie Bridy, "Coding Creativity: Copyright and the Artificially Intelligent Author", *Stanford Technology L Rev* 5 (2012): 1, ¶49; Arthur Miller, "Copyright Protection for Computer Programs, Databases, and Computer-Generated Works: Is Anything New Since CONTU?" *Harvard L. Rev.* 106 (1993): 977, 1042-1072; Pamela Samuelson, "Allocating Ownership Rights in Computer-Generated Works" *University of Pittsburgh L. Rev.* 47(4) (1986): 1185, 1200-1205.

references within the statute and a body of case law predominantly indicate that only humans can be authors under the current law. Specifically, Section 101 of the Copyright Act characterizes anonymous works as those "where no natural person is identified as an author", [103] implicitly leaning towards a human-centric definition of authorship. This aligns with a longstanding legal understanding, backed by U.S. courts, that copyright, as conceived constitutionally, pertains solely to human authors.[104]

The U.S. Supreme Court has consistently reinforced this viewpoint by plainly stating, for instance, that "[a]s a general rule, the author is […] the *person* who translates an idea into a fixed, tangible expression entitled to copyright protection".[105] In *Feist v. Rural*, the court extensively examined the concepts of authorship and originality, linking them to intrinsically human qualities like "creative spark" and "intellectual production".[106] Earlier judicial decisions have supported this human-exclusive approach to copyright as well, emphasizing that it protects "the fruits of intellectual labor" stemming from "the creative powers of the mind",[107] and that copyright law is limited to "original intellectual conceptions of the author".[108]

the case of *Naruto v. Slater* questioned whether a monkey could hold copyright over selfies it had taken. The District Court dismissed the case, stating that nonhuman animals lack statutory standing under the Copyright Act. Even though the case was settled, the Court of Appeals upheld the lower court's decision, arguing that the language used in the Copyright Act implies authorship is a human prerogative.

Further solidifying this position, *Naruto v. Slater* served as a practical test of these principles. In this case, a monkey named Naruto took selfies, which were subsequently published by photographer David Slater in a book by Blurb Inc., attributing copyright ownership to Slater and Wildlife Personalities Ltd.. In 2015, People for the Ethical Treatment of Animals (PETA) filed a lawsuit claiming copyright infringement on Naruto's behalf. This case presented an opportunity for the court to examine if a non-human animal, in this case Naruto, could hold a copyright. The District Court dismissed the case, stating that non-human animals lack the statutory standing required under the Copyright Act.[109] The court further stated that if people acting on Naruto's behalf wanted to argue for animal copyright, they should take the issue up with Congress, not the federal judiciary.[110] The decision was later appealed, and even though the parties reached a settlement, the Court of Appeals chose not to dismiss the case, instead upholding the lower court's ruling. The Appeals Court reasoned that the language of the Copyright Act, specifically words like

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[103] 17 U.S.C. § 101.

[104] See Atilla Kasap, "Copyright and Creative Artificial Intelligence (AI) Systems: A Twenty-First Century Approach to Authorship of AI-Generated Works in the United States", *Wake Forest Intell. Prop. L. J.* 19(4) (2019): 335, 358; Ralph Clifford, "Intellectual Property in the Era of the Creative Computer Program: Will the True Creator Please Stand up", Tulane L. Rev. 71 (1996): 1675, 1682-1686; Timothy Butler, "Can a computer be an author? Copyright aspects of artificial intelligence", (Comm/Ent), A *J. of Communications and Entertainment L.* 4(4) (1981): 707, 733-734; Karl Milde, "Can a Computer Be an "Author" or an "Inventor"?", J. of the Patent Office Soc'y 51 (1969): 378, 391-392.

[105] *Community for Creative Non-Violence v. Reid* (1989) 490 U.S. 730, 737.

[106] *Feist Publications v. Rural Telephone Service* (1990) 499 U.S. 340, 345, 347.

[107] *Trade-Mark Cases* (1879) 100 U.S. 82, 94.

[108] *Burrow-Giles Lithographic Co. v. Sarony* (1884) 111 U.S. 53, 58.

[109] *Naruto v. David Slater* 15-cv-04324-WHO (2016)

[110] ibid.

"children," "grandchildren," "legitimate," "widow," and "widower," implies that copyright protections do not extend to entities that neither marry nor have legally recognized heirs.[111]

The implications of this ruling can logically be extended to include any non-human and AI-generated creative works. In this regard, adding administrative weight to this judicial view, the Third Edition of the Compendium of U.S. Copyright Office Practices has explicitly excluded non-human authorship.[112] The guidance, though non-binding, states that copyright registration is limited to works created by human beings, reaffirming previous court rulings. More specifically, under Section 306, 'The Human Authorship Requirement' limits registration to 'original intellectual conceptions of the author' created by a human being. Section 313.2 of the compendium clarifies that works created by nature, animals, machines, or through mechanical processes without human intervention are not subject to copyright protection.

In at least two instances already, the US Copyright Office has rejected registration of comic books apparently generated via AI, as in the case of Zarya of the Dawn created by Kris Kashtanova via Midjourney and Stephen Thaler's AI-generated painting, "A Recent Entrance to Paradise".[113]In the language used in the letter regarding "Zarya of the Dawn", the U.S. Copyright Office has clarified his position in more detail. The images were generated by the AI system Midjourney, while the arrangement of images, selection, and accompanying text were created by Kashtanova. The Copyright Office granted copyright protection for the text and the compilation, but not for the AI-generated images. First, the Office emphasises human creativity, as seen in the references to the *Urantia* court case, which established that copyright laws are intended to protect creations involving human creativity.[114] The letter states that the images generated by Midjourney are not original works of authorship protected by copyright, as they are produced by a machine without any creative input or intervention from a human author.[115] Second, the letter also highlights the issue of control and the difference between AI-generated works and other computer-based tools such as Adobe Photoshop, which artists control and guide to create their desired images. It argues that users of Midjourney do not have comparable control over the initial image generated, or any final image.[116] Instead, "rather than a tool that Ms. Kashtanova controlled and guided to reach her desired image, Midjourney generates images in an unpredictable way".[117] As a result, the letter concludes that users of Midjourney are not considered "authors" for copyright purposes. Finally, the analogy to a commissioned work of art in the letter further supports the Copyright Office's

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[111] *Naruto v. David Slater*, F.3d 418, 426 (9th Cir, 2018).

[112] U.S. Copyright Office, 'Compendium of U.S. Copyright Office Practices' (3rd edn, U.S. Copyright Office), https://www.copyright.gov/comp3/comp-index.html.

[113] See U. S. Copyright Office, Letter to Mr. Van Lindberg re Zarya of the Dawn (Registration # VAu001480196), February 21, 2023, https://drive.google.com/file/d/1EK7jKVqRz_GP0kJPzOZY_HqKW0n2ML6c/view; U.S. Copyright Office, Letter to Ryan Abbott Re "Second Request for Reconsideration for Refusal to Register a Recent Entrance to Paradise", February 14, 2022, https://www.copyright.gov/rulings-filings/review-board/docs/a-recent-entrance-to-paradise.pdf.

[114] See *Urantia Found. v. Kristen Maaherra*, 114 F.3d 955, 957–59 (9th Cir. 1997) (holding that "some element of *human creativity* must have occurred in order for the Book to be copyrightable" because "it is not *creations* of divine beings that the copyright laws were intended to protect") (emphasis added).

[115] See U. S. Copyright Office, Letter to Mr. Van Lindberg,

[116] ibid., 9.

[117] ibid.

position.[118] It argues that if an individual commissioned a visual artist to create an image based on certain guidelines, the individual would not be considered the author of that image, unless it qualified as a work made for hire. The author would be the visual artist who received those instructions and determined how to express them. In a new "Copyright Registration Guidance: Works Containing Material Generated by Artificial Intelligence", the U.S. Copyright Office crystalizes further the position of the Kashtanova letter by clarifying the copyright status of human arrangement and compilation of AI-generated creativity:

> A human may select or arrange AI-generated material in a sufficiently creative way that "the resulting work as a whole constitutes an original work of authorship." Or an artist may modify material originally generated by AI technology to such a degree that the modifications meet the standard for copyright protection. In these cases, copyright will only protect the human-authored aspects of the work, which are independent of and do not affect the copyright status of the AI-generated material itself.[119]

On August 18, 2023, the U.S. District Court for the District of Columbia, upheld in *Thaler v. Perlmutter* the U.S. Copyright Office's stance that works created entirely by AI are not eligible for copyright protection.[120] The case involved AI scientist Stephen Thaler, who had repeatedly attempted to register a visual art piece titled "A Recent Entrance to Paradise" produced by an AI system he created, known as the "Creativity Machine". Thaler argued that he should own the copyright to the artwork under various legal theories, including the work-for-hire doctrine.[121] He also challenged the Copyright Office's "human authorship" requirement, advocating for AI systems to be recognized as authors when traditional authorship criteria are met, with copyright ownership then transferring to the AI's owner.[122] However, the court sided with the Copyright Office, stressing that "human authorship is a bedrock requirement of copyright".[123] The judgment leaves open questions about the extent of human involvement required for copyright protection, suggesting that future cases may further clarify this issue as AI technology and creative practices continue to evolve.

2.2.1.4. The Chinese Perspective

China stands out as one of the few jurisdictions, and notably the first, where courts have specifically addressed the issue of copyright eligibility for AI-generated creativity. One notable case is *Beijing Feilin Law Firm v. Baidu Corporation*, where the Beijing Internet Court ruled that works solely created by machines are not eligible for copyright protection, emphasizing the necessity of human involvement for such rights.[124] The case revolved around a report disseminated

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[118] ibid., 10.

[119] U.S. Copyright Office, Copyright Registration Guidance: Works Containing Material Generated by Artificial Intelligence, 88 FR 16190, 16192-93, March 16, 2023, https://www.federalregister.gov/documents/2023/03/16/2023-05321/copyright-registration-guidance-works-containing-material-generated-by-artificial-intelligence,

[120] See *Stephen Thaler v. Shira Perlmutter, Register of Copyrights and Director of the United States Copyright Office, et al.*, No. 22-cv-1564 (BAH) (D.D.C. 2023), https://ecf.dcd.uscourts.gov/cgi-bin/show_public_doc?2022cv1564-24.

[121] ibid., 7, 14.

[122] ibid., 3.

[123] ibid., 8.

[124] *Beijing Feilin Law Firm v Baidu Corporation* (26 April 2019) Beijing Internet Court, (2018) Beijing 0491 Minchu No. 239; Kan He, "Feilin v. Baidu: Beijing Internet Court tackles protection of AI/software-generated work and holds

by a Beijing law firm through its official WeChat account. When an anonymous internet user shared the report without authorization, the law firm filed a copyright infringement lawsuit. The contested report was created using Wolters Kluwer China Law & Reference, a legal research software. While the law firm argued that the software merely assisted in the report's creation, the defendants countered that the software generated the entire report autonomously. To resolve this, a report auto-generated by the software—based on keywords provided by the plaintiff's lawyer—was compared to the report in question. The two reports were found to be substantially different, indicating human input in the law firm's report, thus making it eligible for copyright protection under Chinese law. Beyond this specific instance, the court also weighed in on the broader issue of AI-generated works. It asserted that the concept of authorship, to be protected under copyright law, requires a work to be originated by a human. The court did, however, express the belief that users of creative software should receive some form of protection to encourage the purchase and utilization of such software for generating and disseminating works.[125] The judgment, however, stopped short of providing any detailed guidance or recommendations on this particular aspect.

In a subsequent case, *Shenzen Tencent v Yinxun*, the Nanshan District Court in Shenzhen effectively affirmed the earlier Beijing ruling, emphasizing the necessity of human involvement for copyright protection rather than recognizing solely machine-generated creativity.[126] Tencent Technology had developed an AI-based writing assistant called Dreamwriter. In August 2018, Tencent posted an article on its website credited to Dreamwriter, making it clear that the article was the product of their AI technology. The Defendant later republished this article online without Tencent's authorization. In the ensuing copyright infringement lawsuit, Tencent contended that the article was crafted under its oversight, asserting that the company was accountable for its creation and any resultant liabilities. The court sided with Tencent, stating that the article qualified as an original literary work. This was because the content emerged from the data inputs, triggering conditions, and the template and resource configurations chosen by Tencent's operational team. Given that the article's expression was shaped by the deliberate choices and arrangements made by Tencent's human team, the court deemed the AI-generated article as a "work for hire" under Article 11 of the Chinese Copyright Law. The defendant was thus found liable for copyright infringement. Although the court could have interpreted the article as a collaborative intellectual effort between human input and Dreamwriter's operations, the legal protection was explicitly anchored in the contributions made by the human team rather than the AI system.

2.2.2. Originality

Setting aside the human-centric understanding of authorship, the very notion of originality acts as an obstacle to granting copyright protection for creativity generated by AI. Legal texts and precedents generally define originality through an anthropocentric lens, focusing on attributes like self-consciousness. Most legal jurisdictions adopt what is known as a "personality approach" to originality, viewing an original work as an extension or expression of the author's own

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that copyright only vests in works by human authors", *The IPKat*, November 9, 2019, http://ipkitten.blogspot.com/2019/11/feilin-v-baidu-beijing-internet-court.html; Ming Chen, "Beijing Internet Court denies copyright to works created solely by artificial intelligence" *JIPLP* 14(8) (2019): 593.

[125] Rather than the software developer who is already rewarded by a copyright over the software. ibid

[126] See *Shenzhen Tencent v. Yinxun*, Nanshan District People's Court of Shenzhen, Guangdong Province [2019] No. 14010, https://mp.weixin.qq.com/s/jjv7aYT5wDBIdTVWXV6rdQ. See also Kan He, "Another decision on AI-generated works in China: Is it a work of legal entities?", *The IPKat*, January 29, 2020.

personality.[127] This modern understanding has largely marginalized older views that supported 'sweat of the brow' doctrines, which awarded copyright based on the skill, labor, and effort invested in creating a work, irrespective of whether it embodied the author's personality.[128] Consequently, the notion of originality, framed as an expression of self- consciousness or individuality, is theoretically inaccessible to machine-generated creations.[129]

In the European Union, the concept of originality has been harmonised through three key Directives: the Software, Term, and Database Directive. These directives state that a work qualifies as original if it represents "the author's own intellectual creation".[130] The Court of Justice of the European Union (CJEU) subsequently broadened this harmonised concept to encompass all types of copyrighted material. In the *Infopaq* case, the CJEU articulated that originality is expressed "through the choice, sequence, and combination of words," allowing the author to manifest their creativity uniquely.[131] Further clarifying the notion, the *Eva-Maria Painer* decision stated that a work is deemed original—and thus eligible for protection—if it meets three criteria: it must be an intellectual creation of the author; it must reflect the author's personality; and it must demonstrate the author's free and creative choices in its production.[132] These choices provide the work with the author's unique "personal touch".[133] Finally, in the *Football Dataco* case, the CJEU eliminated any lingering support for the "sweat of the brow" doctrine, emphasizing that sheer labor and skill on the part of the author are insufficient for copyright protection unless they exhibit genuine originality.[134]

In the United States, the prevailing approach to defining originality in copyright law is also rooted in the "personality theory", which considers a work to be original if it reflects the creative choices of the author and thereby captures their individual personality,[135] "such as the final product duplicates his conceptions and visions" of what the work should be.[136] Seminal cases like *Burrow-Giles v. Sarony* established that originality emanates from the author's unique creative insights, referring to photographs as copyrightable when originating from the photographer's "unique mental conception".[137] Subsequently, the *Feist v. Rural* decision clarified that mere labor and effort in crafting a work are insufficient criteria for copyright protection; the work must exhibit a

[127] This characterization of originality builds upon Idealist personality theories, according to which intellectual products are manifestations or extensions of the personalities of their creators. See Frosio, "Four Theories", 158-159.

[128] See e.g., *International News Service v Associated Press* 248 U.S. 215 (1918); *Jeweler's Circular Publishing Co. v Keystone Publishing Co.* 281 F. 83, 88 (2nd Cir. 1922); Andreas Rahmatian, "Originality in UK Copyright Law: the Old ''Skill and Labour'' Doctrine Under Pressure", 44 IIC, (2013): 4–34.

[129] The word author itself would bear this meaning on its face as the most accredited etymology of the word would have it deriving from the ancient Greek "αὐτός", which means "self". See Giancarlo Frosio, *Reconciling Copyright with Cumulative Creativity: The Third Paradigm* (Edward Elgar, 2018), 16.

[130] Respectively, Article 1(3), Article 6, and Article 3(1). For a discussion, see Eleonora Rosati, *Originality in EU Copyright - Full Harmonization through Case Law* (Edward Elgar, 2013).

[131] CJEU, C-5/08 *Infopaq*, para 45.

[132] CJEU, C-145/10 *Eva Maria Painer,* para 94.

[133] ibid para 92.

[134] CJEU, C-604/10, *Football Dataco Ltd and Others v Yahoo! UK Ltd and Others*, 2012, ECLI:EU:C:2012:115, para 42.

[135] See *Burrow-Giles Lithographic*, 60-61 (considering the copyrightability of a portrait photograph of Oscar Wilde).

[136] *Lindsay v The Wrecked and Abandoned Vessel R.M.S. Titanic* 52 U.S.P.Q.2d 1609, 1614 (S.D.N.Y. 1999).

[137] *Burrow-Giles Lithographic*, 54-55.

minimum degree of creativity reflective of the author's personality.[138] Following the United States' adherence to the Berne Convention in 1988 and the *Feist* decision in 1991, a more harmonized global view of copyright emerged, emphasizing the personality theory approach to originality.[139] In fact, some legal scholars have argued that U.S. copyright law does not explicitly exclude machine-generated works from protection, citing the low threshold for what constitutes originality.[140] However, the focus should not be on the degree of originality (whether it is high or low). After *Feist*, originality is not only a question of quantum. For works generated by AI, the obstacle is not the amount, but the type of originality required. AI systems, lacking the capability for self-expression, cannot meet the criterion of imbuing their creations with personality, as they possess no personality to express.

In recent years, several primarily common-law jurisdictions, including Australia, India, and the United Kingdom, have shifted towards adopting a personality-centered approach to the concept of originality in copyright law.[141] This move has led these countries to abandon earlier frameworks that emphasized "labor, skill, and effort," commonly known as the "sweat of the brow" doctrines. A few jurisdictions like South Africa and New Zealand continue to adhere to this older view.[142]

Overall, the prevailing interpretation of originality leans heavily on an anthropocentric perspective, asserting that a work can be deemed original only if it serves as an expression of the author's individual personality or “self”. This understanding implicitly assumes that only a sentient, self-aware entity could fulfill such a criterion. Without the element of self-consciousness in the creator, the originality requirement, which hinges on the representation of the author's personality, cannot be satisfied. As a result, under the existing legal framework, AI-generated works fall short of meeting the threshold of originality, unless future advancements in strong AI reach a point where machines gain self-consciousness.

3. The (A)Infringer

The question of whether it is possible for an AI to commit copyright infringement by producing an output that closely resembles or is substantially similar to a copyrighted work is at the center of multiple pending judicial cases. The answer, while straightforward at a glance, is subject to legal intricacies that differ across jurisdictions.

In the United States, copyright infringement is considered a strict liability offense, meaning that fault or intent is not a requirement for liability. For an act to be deemed infringing, three conditions must be met: the act must involve copying rather than independent creation; the resulting copy

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[138] See *Feist Publications,* 362-363.

[139] See Monroe Price and Malla Pollack, “The Author in Copyright: Notes for the Literary critic”, in *The Construction of Authorship: Textual Appropriation in Law and Literature*, eds. Martha Woodmansee and Peter Jaszi (Duke University Press, 1994), 717-720.

[140] Samuelson, “Allocating Ownership Rights in Computer-Generated Works”, 1199-1200. See also Nina Brown, “Artificial Authors: A Case for Copyright in Computer-Generated Works” *Columbia Science and Tech. L. Rev.* 9 (2019): 1, 24-27; Margot Kaminski, “Authorship, Disrupted: AI Authors in Copyright and First Amendment Law”, 51 *UC Davis L. Rev. (*017): 589, 601.

[141] See e.g., *IceTV,* 43; *Eastern Book Co. & Ors v D.B. Modak & Anr* (2008) 1 SCC 1; *Temple Island Collections v New English Teas (No. 2)* [2012] EWPCC 1; Rahmatian, “Originality in UK Copyright Law”, 4–34.

[142] See e.g., *Appleton v Harnischfeger Corp.* (1995) 2 SA 247 (AD), [43]– [44]; *Henkel KgaA v. Holdfast* [2006] NZSC 102, [2007] 1 NZLR 577 [37] (NZ).

must be fixed in a tangible medium; and the act must involve the improper appropriation of protected material. While the requirement of being 'fixed in a tangible medium' is more specific to U.S. law, the conditions concerning 'copying' and 'improper appropriation' are broadly relevant in many legal jurisdictions. Copying can occur in several forms: mechanical reproduction, creating a work substantially similar to a copyrighted one, or translating the original work to another medium. Importantly, copyright infringement does not require intent.[143] Case law suggests that even 'innocent' or 'subconscious' copying is no defense,[144] if access to the original work can be proven[145] or the similarities between the unusual aspects of the two works are sufficiently striking.[146] As for 'improper appropriation,' this depends on the character and extent of the material taken. Copyright infringement can occur either by copying the entire work, by taking significant qualitative or quantitative parts of it, or by producing a work that is substantially similar in essence.[147] Therefore, if the AI's generated output borrows extensively from a copyrighted work or is substantially similar to it, it would qualify as infringement. If the use of the copyrighted material is transformative, however, one must consider whether such use falls under the fair use doctrine.[148] Generally speaking, transformative use is protected by fair use provisions and does not constitute copyright infringement.[149] Therefore, whether the AI-generated images qualify as fair use would hinge on how transformative they are with respect to the original copyrighted works.

The discussion above serves to address a crucial question: Can AI-generated outputs infringe upon the copyrights of the images, people, places, and objects used for training the models? The short answer is “yes”, but only if the protected content appears in the output in an identical or substantially similar form. Copyright infringement does not occur if the original content merely serves as an inspiration and is either minimally replicated in the output[150] or used in a transformative way.[151] In the ongoing cases, both Stability AI and Midjourney assert that proof of actual infringement or substantial similarity in the generated outputs has yet to be conclusively demonstrated.[152] Midjourney, in particular, challenges the notion that just because the outputs are “derived from” the images used for training their model, they should automatically be classified as derivative works.[153] They contend that the plaintiffs are misleadingly conflating the legal

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[143] See *Fisher v. Dillingham,* 298 F. 145, 148 (SDNY 1924) (“it is no excuse that memory played a trick”).

[144] See *Bright Tunes v. Harrison*, 420 F. Supp. 177, 181 (SDNY 1976).

[145] See *Three Boys Music v. Michael Bolton*, 212 F.3d 477 (9th Cir. 2000).

[146] See e.g., *Selle v. Gibb*, 741 F.2d 896 (7th Cir. 1984).

[147] See e.g., Daniel Gervais, “Improper Appropriation”, *Lewis and Clark L. Rev.* 23(2) (2019): 600; Oren Bracha, “Not de Minimis: (Improper) Appropriation in Copyright”, *American U. L. Rev.* 68 (2018): 139.

[148] See e.g., 17 USC § 107; Casey Newton, “How DALL-E could power a creative revolution”, *The Verge*, June 10, 2022, https://www.theverge.com/23162454/openai-dall-e-image-generation-tool-creative-revolution.

[149] See Jiarui Liu, “An Empirical Study of Transformative Use in Copyright Law”, *Stanford Tech. L. Rev.* 22 (2019): 163, 166; Pierre Leval, “Toward a Fair Use Standard”, *Harvard L. Rev.* 103, (1990): 1105, 1111.

[150] See e.g., Gervais, “Improper Appropriation”, 600; Bracha, “Not de Minimis”, 139.

[151] See Liu, “An Empirical Study of Transformative Use in Copyright Law”, 166; Leval, “Toward a Fair Use Standard”, 1111.

[152] See *Andersen et al. v. Stability Ai et al.*, Case no 3:23-cv-00201-WHO (N.D. Cal., April 18, 2023) (Stability AI’s Amended Mot to Dismiss), https://www.courtlistener.com/docket/66732129/andersen-v-stability-ai-ltd.

[153] Cf. Daniel Gervais, “AI Derivatives: the Application to the Derivative Work Right to Literary and Artistic Productions of AI Machines”, *Seton Hall L. Rev.* 52(4) (2022): 1111, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4022665.

copyright concept of a "derivative work" with a terminology they have devised to describe the output created by their users.

3.1. Protection of Style

Artists have expressed significant concern over the proliferation of AI-generated art that closely mimics their unique styles. These anxieties are not just abstract worries; they're directly related to ongoing legal disputes involving AI platforms such as Dall-E, Stable Diffusion, and Stability AI. Users of the AI image-generating platforms are accused of creating works that imitate the style of specific artists, including those who are Plaintiffs in the lawsuits.[154] These users would employ prompts containing the artists' names to generate outputs, misleadingly suggesting that the works are original creations by those named artists. Referred to as 'Imposters' in the context of the case, these users manipulate the AI to lean heavily on the stylistic elements of the mentioned artists, producing images that closely resemble those artists' unique styles.[155]

Creators are alarmed by the extent to which AI-generated images, designed to imitate their styles, are flooding online platforms and even overshadowing their original works in search results.[156] Greg Rutkowski, the most popular artist used to generate AI images and well-known for producing fantastical high-quality images, showed concern for the mass copying of individual styles by AI by noting "right now, when you type in my name, you see more work from the AI than work that I have done myself, which is terrifying for me. How long till the AI floods my results and is indistinguishable from my works?"[157] This phenomenon raises pressing questions that are at the heart of current litigation: to what extent can or should generative AI be held accountable for infringing on artists' styles?

However, from a copyright law perspective,[158] these concerns may be somewhat misplaced. Copyright law traditionally protects the specific expression of ideas, not the general style or techniques used to achieve that expression. In other words, while an artist's unique manner of expression is noteworthy, it isn't what copyright law aims to protect. Instead, copyright would protect the specific expression of that style in a particular work. Courts have consistently held that style is just one element of artistic expression; infringement only occurs when there is substantial similarity between the specific expressions of two works.[159] In *Jewelry* 10, the court explains that

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[154] See Andersen et al. (Complaint), §5-6

[155] ibid., §171-176.

[156] See, e.g., Johnston, "Comic Book Creators React to AI Artificial Intelligence Art Explosion"; Baio, "Invasive Diffusion".

[157] Matt Growcoot, "AI Image Generator Stable Diffusion Releases Controversial New Version", *PetaPixel*, (, November 28, 2022, https://petapixel.com/2022/11/28/ai-image-generator-stable-diffusion-releases-controversial-new-version. See also Melissa Heikkilä, "This artist is dominating AI-generated art. And he's not happy about it. Greg Rutkowski is a more popular prompt than Picasso", *MIT Technology Rev.*, September 16, 2022, https://www.technologyreview.com/2022/09/16/1059598/this-artist-is-dominating-ai-generated-art-and-hes-not-happy-about-it.

[158] This might not be the case in all jurisdictions. There is, for example, at least one case in Italy—finally settled—where a Milanese court found an album cover of Roger Waters apparently infringing upon the "cancellation" technique of Emilio Isgrò. See e.g., Gabriele Antonucci, "Roger Waters – Emilio Isgrò: perché è plagio l'artwork dell'album", *Panorama*, July 26, 2017, https://www.panorama.it/lifestyle/musica/roger-waters-e-davvero-plagio-lartwork-di-life-really-want (in Italian only).

[159] See, e.g., *Steinberg v. Columbia Pictures Industries, Inc.*, 663 F. Supp. 706, 706-716 (SDNY, 1987); *Cummins v Vella* [2002] FCAFC 218.

infringement will be found in "a detailed copying which takes not only the stylistic idea but the manner or details of execution".[160] A ruling in the case Dave *Grossman Designs v. Bortin* clarified that an artist can use another's style without infringing on copyright, as long as they don't substantially copy the specific expression of the original work. For example, the Court says, "Picasso may be entitled to a copyright on his portrait of three women painted in his Cubist motif", but other artists "may paint a picture of any subject in the Cubist motif, including a portrait of three women, and not violate Picasso's copyright so long as the second artist does not substantially copy Picasso's specific expression of his idea".[161]).

In summary, if an AI generates an artwork that merely echoes the style of an existing artist—without closely replicating specific, copyright-protected elements of that artist's work—it would generally not be considered an infringement under current copyright law. Copyright protection encompasses a variety of factors like composition, framing, content, style, narrative, and artistic intent that together make a work original and a representation of the author's personality.[162] Isolating one aspect, like style, typically does not meet the threshold for legal protection.

3.2. Who is infringing? Asking the Right Questions

A recent policy statement on the matter from Japan sheds some light on the distinction between potentially infringing input and infringing output and seems to suggest that AI-generated art is not inherently in violation of existing copyright laws. According to the Agency for Cultural Affairs of Japan, there are two key phases in the creation of AI-generated art: the learning and research stage, and the stage of image and artwork generation.[163] The ACA's document titled document titled "About the relationship between AI and copyright" claims that copyrighted works can be used without permission during the AI training phase, since these works would be used for non-commercial purposes.[164] However, the commercial production and sale of AI-generated art would be subject to standard copyright infringement laws.[165] If an AI-generated work is found to be derivative of an existing copyrighted piece, legal ramifications including damages and criminal charges could apply. This approach upholds the conventional mechanisms for copyright protection: if you believe your copyrighted work has been infringed upon by an AI-generated piece, you can take legal action in much the same way you would for human-created art. In essence, the standard legal procedures remain applicable. If a lawsuit clearly demonstrates that a specific piece of AI-generated art is a direct copy of existing copyrighted work, then the original artist can claim damages. On the other hand, if the claim is based merely on a subjective sense of similarity without concrete evidence, the lawsuit is likely to fail, just as it would with disputes over non-AI art.

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[160] See *Jewelry 10, Inc. v. Elegance Trading Co.*, 1991 WL 144151 (S.D.N.Y. 1991).

[161] *Dave Grossman Designs v. Bortin*, 347 F. Supp. 1150, 1156 (N.D. Ill. 1972).

[162] See, e.g., Steinberg, 706-716; *Malden Mills, Inc. v. Regency Mills, Inc.*, 626 F.2d 1112 (2d Cir. 1980) ("The two designs are of such likeness with regard to subject matter, style of representation, shading, composition, relative size and placement of components, and mood as to obviously substantially similar"); *Burrow-Giles Lithographic,* 53ff. See also *Feist*; CJEU, C-145/10, para 99.

[163] See Daniel Bueno, "AI Art Will Be Subject to Copyright Infringement in Japan", *Siliconera*, June 6, 2023, https://www.siliconera.com/ai-art-will-be-subject-to-copyright-infringement-in-japan

[164] ibid.

[165] ibid.

Nevertheless, given that the machine generates its output based on specific user instructions, it seems reasonable to argue that responsibility for violations stemming from the output should predominantly fall on the user. In essence, these infringements would not have occurred without the user's active role in eliciting the contentious material. Therefore, it's reasonable to categorize the machine as the "primary infringer," with the user assuming "secondary liability" for instigating the infringement through their directives. The rationale behind this is that the machine, trained to generate a potentially infinite array of outputs by its developers, only produces the infringing material as a direct result of the user's targeted instructions. Without these guidelines, the contentious output would merely exist as one possibility among countless others. A generative AI model has a range of possible outputs based on its training data and algorithms. The user's specific instructions act as a filter, steering the AI towards generating content that is substantially similar to copyrighted material. In this sense, the user can be seen as the enabler or inducer of the infringement, making them secondarily liable. Their specific instructions are what transform a potential infringement into an actual one, from among the myriad outputs the AI could generate based on its training. AI's capabilities extend to a virtually limitless array of possible outputs. Without the user's specific directives, the machine's potential to generate infringing content remains just that—potential. It's the user's intent and actions that crystallize this potential into actual, actionable infringement. Therefore, the liability for infringement could be more closely linked to the queries posed to the machine, rather than the machine's responses.

The advent of AI technology does prompt a rethinking of traditional frameworks for copyright infringement and courts might need to change the litmus test for infringement.[166] Traditionally, the focus has been largely on the end-product, the “answer” that Generative AI provides to users prompts. The entity that produced a piece of work substantially similar to an existing copyrighted work would be considered liable for infringement. However, with AI stepping into the creative process, the lines become blurred, and the focus could potentially shift to the questions, or directives, that prompt the creation of such content. Generative AI compels us to rethink the concept of copyright infringement from a product-centric to a process-centric viewpoint. The 'questions' leading to the final work may become as, if not more, important than the work itself in determining liability.

Legal systems will need to adapt to accommodate this paradigm shift, making room for nuanced dialogues about intentionality, specificity, and the interactive nature of modern content creation. First, courts might shift to intentionality. In a world of machine-generated content, a key criterion for infringement might become the intent behind the directive given to the machine. Here, the focus wouldn't be just on the similarity between the copyrighted work and the new work, but also on the intent of the user who provided the instructions to the machine. Did they knowingly or purposefully induce the AI to create something similar to an existing copyrighted work? Second, the context in which a directive is given may also gain importance. For instance, if a user asks an AI to "create something like X," where X is a copyrighted work, then even if the machine's output isn't substantially similar, the intention to infringe was clearly present. Third, courts might need to consider more carefully the interactive nature of AI. Traditional tests for infringement might not work well for AI because AI is interactive. The process of creation often involves back-and-forth interaction between the user and the machine. This makes each directive part of an ongoing

[166] See Mark Lemley, “How Generative AI Turns Copyright Law on its Head” SSRN Research Paper no 4517702/23, July 21, 2023, https://ssrn.com/abstract=4517702 (noting that “we may need to throw out our test for infringement, or at least apply it in fundamentally different ways”).

dialogue, which cumulatively leads to the final product. Tracking this dialogue could become important in determining infringement.

Finally, given the shift towards questioning, developers might have a role to play here, too, and they might be liable under certain circumstances. They might be required to implement pre-emptive measures and could incorporate safeguards into the AI to flag potentially infringing directives, or to require additional confirmations from the user, making them more aware of the implications of their directives. In this very regard, In light of the growing concern that learned conditional generative models may output samples that are substantially similar to some copyrighted data that was in their training set, researchers are "trying to come up with a mathematical definition guaranteeing that the output of a generative model does not infringe on a copyrighted work, even if that work is present in its training set."[167] In the absence of these technical safeguards, courts may extend liability to Generative AI platforms and their developers for producing outputs that infringe on copyrights. In the same fashion, courts might consider that if the developers provide a platform that allows users to generate works "in the style of a certain artist", and these generated works closely resemble the original artists' styles, then a preliminary claim for copyright infringement appears viable. However, to truly assess the legitimacy of any infringement claim, a detailed comparison between the plaintiffs' original works and the alleged derivative works produced by the platforms is essential. To unequivocally address these pressing concerns, Stability AI has actually taken the deliberate step of disabling the feature that allows the generation of images in the styles of specific artists in its latest update.[168]

CONCLUSIONS

In this chapter, we delved deep into the intricate interplay between copyright law and the rapidly advancing field of Generative AI. While the law has yet to fully adapt to these technological advances, ongoing litigation against generative AI platforms stands as a testament to the pressing need for legal clarification. The courts have a pivotal role to play in shaping these norms. Amidst the legal grey areas and pending litigation, judicial decisions will set important precedents that could either stimulate or stifle innovation in this arena.

Courts ought to approach Generative AI with the nuanced understanding that it is a dual-use technology, holding the potential for both significant societal benefits and copyright infringement risks. Historically, such technologies aren't strictly regulated unless they explicitly encourage unlawful activity.[169] As with online platform intermediary liability,[170] the focus may be best placed on platforms that negligently enable copyright violations. If a platform has already taken steps to

[167] See Nikhil Vyas, Sham Kakade, Boaz Barak, "On Provable Copyright Protection for Generative Models", arXiv:2302.10870 [cs.LG] (2023), https://arxiv.org/abs/2302.10870.

[168] See Stable Diffusion Version 2, https://github.com/Stability-AI/stablediffusion. See also Matthias Bastian, "Stable Diffusion v2 removes NSFW images and causes protests", *The Decoder*, November 26, 2022, https://the-decoder.com/stable-diffusion-v2-removes-nude-images-and-causes-protests.

[169] See, e.g., *Sony*, 417; Reiner Kraakman, "Gatekeepers: the Anatomy of a Third-Party Enforcement Strategy", *Journal of Law, Economics and Organization* 2(1) (1986): 53.

[170] See, e.g., The Digital Millennium Copyright Act of 1998, 17 U.S.C. § 512; Directive 2000/31/EC of the European Parliament and of the Council of 8 June 2000 on certain legal aspects of information society services, in particular electronic commerce, in the Internal Market, 2000 O.J. (L 178) 1-16, art 12-15; Regulation of the European Parliament and of the Council of 19 October 2022 on a Single Market for Digital Services and amending Directive 2000/31/EC (Digital Services Act), 2022 OJ L 277/1.

prevent misuse, the responsibility should shift to the user and follow traditional copyright infringement dynamics. In essence, the courts have the opportunity to create a balanced regulatory approach, safeguarding artists' rights while not stifling technological innovation. Their decisions will set the tone for the ethical and legal use of Generative AI moving forward.